Aayush Gnawali

AI/ML • Backend • DevOps

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Summary

Final-year BE-IT student with practical experience across **AI/ML, backend engineering, and DevOps**. Experienced in developing machine learning applications and RAG-based systems, building Python backend services, and working with containerization, cloud infrastructure, and CI/CD workflows. Interested in applying AI/ML to real-world problems while developing reliable and production-oriented software systems.

Experience

Backend and QA Intern **May 2026 – August 2026**
E-Digital Nepal Pvt. Ltd. *On-site*

- Developed and maintained backend services using **Python and PostgreSQL** for APIs, authentication, role-based access control, and secure data management.
- Performed **manual testing** of backend and application features to identify functional issues and verify expected system behavior.
- Learned and applied **automated testing using pytest and Selenium** to improve test coverage and understand automated testing workflows.
- Contributed to admin tools, dynamic form management, user tracking, and other platform features while collaborating with the engineering team.

DevOps Intern **2026**
Lego Tech Pvt. Ltd. *Remote*

- Worked with **Linux, Bash, Docker, Docker Compose, Git, and CI/CD** across development and deployment workflows.
- Containerized services and supported deployment and environment configuration tasks using Docker and YAML-based configuration.
- Gained practical experience with **AWS EC2**, cloud infrastructure, deployment fundamentals, and distributed engineering workflows.

Selected Projects

Portfolio RAG Assistant *Live Demo*

- Built a **Retrieval-Augmented Generation (RAG)** assistant that retrieves relevant evidence from structured portfolio content before generating responses.
- Implemented document chunking, embeddings, vector retrieval, and grounded LLM response generation.
- Applied public-only document filtering, prompt isolation, output validation, and evidence-based abstention to improve reliability and reduce hallucinations.

House Price Prediction Web App *Live Demo*

- Developed and deployed a **machine learning regression application** for interactive house-price prediction.
- Implemented data preprocessing, feature engineering, model inference, and visualization as part of an end-to-end ML workflow.
- Converted a notebook-based model into a live application capable of providing real-time predictions.

Rule-Based Self-Healing Web Automation System

- Developed a self-healing automation system that detects UI changes and recovers failed browser workflows using rule-based decision logic.
- Designed browser-state tracking and recovery mechanisms to handle selector changes and dynamic application updates.

HippyHub E-Commerce Website *GitHub Repository*

- Built a full-stack application using **React, FastAPI, and PostgreSQL**, with Docker-based service orchestration and deployment.

Technical Skills

AI / ML	PyTorch, Scikit-learn, Pandas, NumPy, Deep Learning, Machine Learning, RAG, Semantic Search, Embeddings
Backend	Python, FastAPI, REST APIs, Async SQLAlchemy, Pydantic, JWT Authentication, PostgreSQL
DevOps	Docker, Docker Compose, Linux, Bash, Git, GitHub, CI/CD, AWS
ML / Application	Model Deployment, Data Preprocessing, Feature Engineering, Streamlit
Testing	pytest, Selenium, Manual Testing
Tools	GitHub Actions, MySQL, Virtual Environments

Education

B.E. in Information Technology **2022 – Present**
Everest Engineering College, Pokhara University *8th Semester*

Higher Secondary Education (+2 Science) **2020 – 2022**
Trinity International College

Certifications

- **AI/ML Workshop** – ITSNP, 2025: Neural network architectures and implementation, including CNNs.
- **DevOps and Cloud Computing Training** – Upskill Nepal, 2026.
- **Robotics Training** – Everest Engineering College, 2025.

Achievements

- **Best Minor Project** – Everest Engineering College, 2025.
- **Chief Content Writer** – Hult Prize Organizing Committee, Everest Engineering College, 2025; team ranked in the Top 25% in South Asia.